

Uncertainty and Model Checking for Atmospheric Inversions

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Model

What we know, and what we want to know:

$\mathbf{y}$: Observed measurements dimension M

$\mathbf{x}$: Unobserved flux dimension N

Hyperparameters and data generating process:

$\mathbf{H}$: Footprints dimension $M \times N$

$\mathbf{B}$: Prior covariance dimension $N \times N$

$\mathbf{R}$: Observation error dimension $M \times M$

$\mathbf{m}$: Prior inventory dimension N

$\mathbf{y} \sim \mathcal{N}(\mathbf{m}, \mathbf{B})$ Actual fluxes are smooth, centered at inventory

$\mathbf{x}|\mathbf{y} \sim \mathcal{N}(\mathbf{H}\mathbf{y}, \mathbf{R})$ The footprints transport $\mathbf{y}$, then observe with some error

Inversions estimate $\hat{\mathbf{x}} = \mathbb{E}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}[\mathbf{x}]$ at great expense!

Would also like, without a ton of extra computation:

- Measurements of posterior uncertainty. (E.g., how large is $\text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(\mathbf{x})$?)
- Measurements of prior sensitivity (E.g., what if the length scale of $\mathbf{B}$ changed?)
- Measurements of data influence (E.g., how much is my data, and how much my prior?)

Data generating process:

$\mathbf{y} \sim \mathcal{N}(\mathbf{m}, \mathbf{B})$ Actual fluxes are smooth, centered at inventory

$\mathbf{x}|\mathbf{y} \sim \mathcal{N}(\mathbf{H}\mathbf{y}, \mathbf{R})$ The footprints transport $\mathbf{y}$, observed with some error

Inference: $\mathcal{P}(\mathbf{y}, \mathbf{x})$ is Gaussian $\Rightarrow \mathcal{P}(\mathbf{x}|\mathbf{y})$ is Gaussian.

$$\mathcal{P}(\mathbf{x}|\mathbf{y}) = \mathcal{N}(\hat{\mathbf{x}}, \hat{\mathbf{V}}) \quad \text{where}$$

$$\hat{\mathbf{x}} = \mathbf{m} + (\mathbf{HB})^\top (\mathbf{HBH}^\top + \mathbf{R})^{-1} (\mathbf{y} - \mathbf{m})$$

$$\hat{\mathbf{V}} = \mathbf{B} - (\mathbf{HB})^\top (\mathbf{HBH}^\top + \mathbf{R})^{-1} (\mathbf{HB})$$

Inversions estimate $\hat{\mathbf{x}} = \mathbb{E}_{\mathcal{P}(\mathbf{x}|\mathbf{y})} [\mathbf{x}]$. This is slow because of

- Forming $\mathbf{H}$ and $\mathbf{B}$ in memory
- Computing $\mathbf{HB}$ and $\mathbf{HBH}^\top$
- Inverting $(\mathbf{HBH}^\top + \mathbf{R})$

Sequoia has sped these up, and also saved to disk what she can.

Assume we can re-use these computations. How can we use them?

$$\mathcal{P}(\mathbf{x}|\mathbf{y}) = \mathcal{N}(\hat{\mathbf{x}}, \hat{\mathbf{V}}) \text{ where } \hat{\mathbf{V}} \text{ is huge } (N \times N)$$

Obviously we can't compute $\hat{\mathbf{V}}$. But just as obviously don't actually want all of $\hat{\mathbf{V}}$!

Key idea: We actually want to know is not $\mathbf{x}$, but linear combinations of $\mathbf{x}$.

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Key idea: We actually want to know is not $\mathbf{x}$, but linear combinations of $\mathbf{x}$.

Example: Want s , the total emissions in hour t , summed over all spatial locations.

Let $\mathbf{a}$ be an N -vector with:

$$\mathbf{a}_n = \begin{cases} 1 & \text{If observation } n \text{ was at time } t \\ 0 & \text{otherwise} \end{cases}$$

Then

$$s = \sum_{n=1}^N \mathbf{a}_n \mathbf{x}_n = \mathbf{a}^\top \mathbf{x} \Rightarrow$$

$$\text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(s) = \text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(\mathbf{a}^\top \mathbf{x}) = \mathbf{a}^\top \text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(\mathbf{x}) \mathbf{a}$$

The covariance of the scalar s is a linear combination of stuff we computed for $\hat{\mathbf{x}}$:

$$\text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(s) = \mathbf{a}^\top \underbrace{\mathbf{B}}_{\text{done}} \mathbf{a} - \underbrace{(\underbrace{\mathbf{HB}}_{\text{done}} \mathbf{a})^\top}_{\text{done}} \underbrace{(\mathbf{HBH}^\top + \mathbf{R})^{-1}}_{\text{done}} \underbrace{(\mathbf{HB} \mathbf{a})}_{\text{done}}$$

$$\mathcal{P}(\mathbf{x}|\mathbf{y}) = \mathcal{N}(\hat{\mathbf{x}}, \hat{\mathbf{V}}) \text{ where } \hat{\mathbf{V}} \text{ is huge } (N \times N)$$

Obviously we can't compute $\hat{\mathbf{V}}$. But just as obviously don't actually want all of $\hat{\mathbf{V}}$!

Key idea: We actually want to know is not $\mathbf{x}$, but linear combinations of $\mathbf{x}$.

Example: Want $\mathbf{s}$, the total emissions in hours $t = 1, \dots, 24$, summed over all spatial locations.

Let $\mathbf{A}$ be an $24 \times N$ matrix with:

$$\mathbf{A}_{tn} = \begin{cases} 1 & \text{If observation } n \text{ was at time } t \\ 0 & \text{otherwise} \end{cases}$$

Then

$$\mathbf{s} = \mathbf{A}\mathbf{x} \Rightarrow$$

$$\text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(\mathbf{s}) = \mathbf{A}\text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(\mathbf{x}) \mathbf{A}^\top$$

The covariance of the vector $\mathbf{s}$ is a linear combination of stuff we already computed.

$$\mathcal{P}(\mathbf{x}|\mathbf{y}) = \mathcal{N}(\hat{\mathbf{x}}, \hat{\mathbf{V}}) \text{ where } \hat{\mathbf{V}} \text{ is huge } (N \times N)$$

Obviously we can't compute $\hat{\mathbf{V}}$. But just as obviously don't actually want all of $\hat{\mathbf{V}}$!

Key idea: We actually want to know is not $\mathbf{x}$, but linear combinations of $\mathbf{x}$.

Extreme example: Want t^* , the hour of the day where the total flux is largest.

The value t^* is not a linear combination of $\mathbf{x}$. But we just showed that

$$\mathcal{P}(\mathbf{s}|\mathbf{y}) = \mathcal{N}(\mathbf{A}\hat{\mathbf{x}}, \mathbf{A}\hat{\mathbf{V}}\mathbf{A}^\top),$$

and everything is computable. So we can:

- Draw $\mathbf{s}_1, \dots, \mathbf{s}_K$ from the multivariate normal
- Compute $t_k^* = \operatorname{argmax}_t \mathbf{s}_{kt}$
- Estimate $\operatorname{Var}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(t^*) \approx \frac{1}{K} \sum_{k=1}^K (t_k^* - \bar{t}^*)^2$

The covariance of the vector t^* easily estimated from stuff we already computed.

The prior $\mathbf{B}$ depends on quantities like ρ , an assumed lengthscale. So we might write

$$\hat{\mathbf{x}}(\rho) = \mathbf{m} + (\mathbf{HB}(\rho))^{\top} (\mathbf{HB}(\rho)\mathbf{H}^{\top} + \mathbf{R})^{-1} (\mathbf{y} - \mathbf{m})$$

Problem:

- Re-computing $\hat{\mathbf{x}}(\rho)$ for a new lengthscale is expensive (a whole new problem).
- There actually can be lots of hyperparameters, and you don't know which matter.

Idea: Starting from a solution at the initial guess ρ_0 , form a Taylor series expansion

$$\hat{\mathbf{x}}(\rho) \approx \hat{\mathbf{x}} + \frac{\partial \hat{\mathbf{x}}(\rho)}{\partial \rho} (\rho - \rho_0)$$

$$\frac{\partial}{\partial \rho} (\mathbf{HB}(\rho)\mathbf{H}^{\top} + \mathbf{R})^{-1} = - \underbrace{(\mathbf{HBH}^{\top} + \mathbf{R})^{-1}}_{\text{done}} \mathbf{H} \frac{\partial \mathbf{B}(\rho)}{\partial \rho} \mathbf{H}^{\top} \underbrace{(\mathbf{HBH}^{\top} + \mathbf{R})^{-1}}_{\text{done}}$$

The Taylor series can be easily estimated from stuff we already computed.

Problem: In inversion problems, the prior is informative, and the model does smoothing.

How is your data being used? What patterns are possible to identify?

Example question: Can I detect emissions from road pixels?

Idea: Let $\mathbf{r}$ be an N -vector encoding road pixels.

$$\mathbf{r}_n = \begin{cases} 1 & \text{If observation } n \text{ was a road pixel} \\ 0 & \text{otherwise} \end{cases}$$

Suppose we had observed not $\mathbf{y}$, but $\mathbf{y}' = \mathbf{y} + \epsilon \mathbf{r}$. That is, $\mathbf{y}'$ is a little more “roady” than $\mathbf{y}$.

Then, suppose I used $\mathbf{y}'$ to compute $\mathbf{x}'$. How much more “roady” is $\mathbf{x}'$ than $\mathbf{x}$?

$$\begin{aligned} \frac{\text{Inferred road pixel change}}{\text{Actual road pixel change}} &= \frac{\mathbf{r}^\top (\mathbf{x}' - \mathbf{x})}{\mathbf{r}^\top (\mathbf{y}' - \mathbf{y})} \\ &= \epsilon^{-1} \frac{\mathbf{r}^\top (\mathbf{x}' - \mathbf{x})}{\mathbf{r}^\top \mathbf{r}} \\ &= \frac{\mathbf{r}^\top \overbrace{(\mathbf{HB})^\top (\mathbf{HBH}^\top + \mathbf{R})^{-1}}^{\text{done}} \mathbf{r}}{\mathbf{r}^\top \mathbf{r}} \end{aligned}$$

This is just the “gain matrix” queried for a particular output.

The data influence can be easily estimated from stuff we already computed.

